

Ethan Niv

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Robotics, path planning and control engineer with experience deploying real-time, hardware–software systems

EXPERIENCE

Robotics Graduate Teaching Assistant - Mechatronics, Dynamics, Math for Robotics and Foundations of Robotics, Sep 2025 – May 2026 NYU Tandon School of Engineering, New York City, United States

- Supported graduate-level robotics coursework covering kinematics, state-space control, estimation (Kalman / BLUE / SVD), and system modeling

Software Engineer, June 2021 – December 2023

Zalirian LTD., Tel Aviv, Israel

- Designed and implemented a production-grade, real-time edge control and perception system, integrating motors, industrial cameras, and external hardware SDKs
- Built high-performance, maintainable software architecture, leveraging multithreading, multiprocessing, design patterns, and C++/Python interoperability (Pybind), with robust logging and telemetry for real-time monitoring and post-deployment support.
- Implemented a Motion Planning algorithm to assure movement efficiency and safety in a 5DoF system with physical limitations and obstacles, thereby improving the movement efficiency and safety.

Software and Hardware Developer, October 2018 – April 2021

Israeli Air Force, Israel

- Spearheaded end-to-end field deployments of mission-critical hardware/software systems, overseeing physical installation, sensor integration, and system validation in live environments.
 - Reverse-engineered and modernized legacy airborne and sensor systems, integrating new hardware with existing infrastructure.
 - Interfaced with complex sensor and data acquisition systems, applying signal processing to evaluate system accuracy, reliability, and safety.
 - Performed system-level fault analysis and debugging to support flight investigations and operational readiness.
- Delivered solutions in environments requiring high reliability, rapid diagnosis, and clear technical documentation.

SKILLS

Software: Python, C++, ROS2, GIT, Pybind, LabVIEW, OpenCV, PyQt, MatLab, Simulink, Nav2, Gazebo, Mujoco, Docker

Technical: Motion Planning, Control Theory, Signal Processing, Sensor and Actuator Integration, Localization and Navigation, EKF, Particle Filtering, Project Management, Client-Development, Requirements analysis, Device Drivers & SDK Integration

PROJECTS

M.Sc Final Project: Designed and implemented a collision mitigation strategy for autonomous vehicles in case of imminent collision. Integrated into the Nav2 library, and implemented a multi-stage MPPI controller, created custom plugins for calculating the surface normals of obstacles, created a custom critic for the controller, designed and created replicable testing scenarios in Gazebo in spite of the randomness of the algorithm. Created a docker container for replicability of the system.

M.Sc Mechatronics Projects: Programmed a SCARA robot arm that can reproduce intricate sketches using RPi, Arduino and ArUco markers • Simulator for forward and inverse kinematics of a 6DOF robotic arm • Path Planning on UR16e robotic arm in Gazebo using ROS2 • Designed and implemented MPC, MPPI controllers for a simulated drone / robotic arm • Designed a full AMR simulation from dynamics, to motion planning and multiple controllers, using a modular architecture for switching between planners and controllers • Implemented state estimation algorithms using Extended Kalman Filters(EKF) and Particle Filters (PF), in conjunction with either a separate camera or a lidar sensor, given a low quality single encoder to achieve robust and accurate localization on a robotic vehicle

B.S Final Project: Designing and implementing a self-balancing quadrotor drone control system.

EDUCATION

New York University Tandon School of Engineering Graduated May 2026 M.S in Mechatronics and Robotics - GPA 3.96

Tel Aviv University Graduated June 2020 B.S. in Electrical Engineering and Computer Science